

Where every new part goes

Four views of the stock coach, with the new hardware placed in it: the floor plan, a side section, the underbody and the roof. Each new part has a number that matches the schedule below, which says where it goes, what it takes the place of, how it mounts and what it connects to. Positions are inches forward of the rear wall. They come from the model and get confirmed with a tape measure.

INSIDE THE CABIN • Power centre in the driver-side lounge seat base: both packs stacked at the rear end, the RV5 at the front end, transfer switch, surge protector, disconnect and PV breakers on the front panel (1–3, 11, 12) • Epad on the wall above the lounge (4) • Washer in the curbside corner at the forward end of the lift bed, against the pantry (13) • Factory desk untouched	ROOF • Six panels in one crosswise row, split around the AC (6) • Combiner and cable gland (7, 5) • Chill Cube swaps into the existing opening (8) • Starlink Mini beside it (9)	UNDERNEATH AND CAB • Nothing new hangs under the coach • Two Orions under the passenger swivel seat, about 6 ft from the packs (10)	NOT CONFIRMED YET • Height under the lowered bed at the washer corner (needs at least 33.7 in) • The other fit checks listed below
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■ Bluetti / 48 V DC

■ Solar

■ Alternator charging

■ 120 V AC

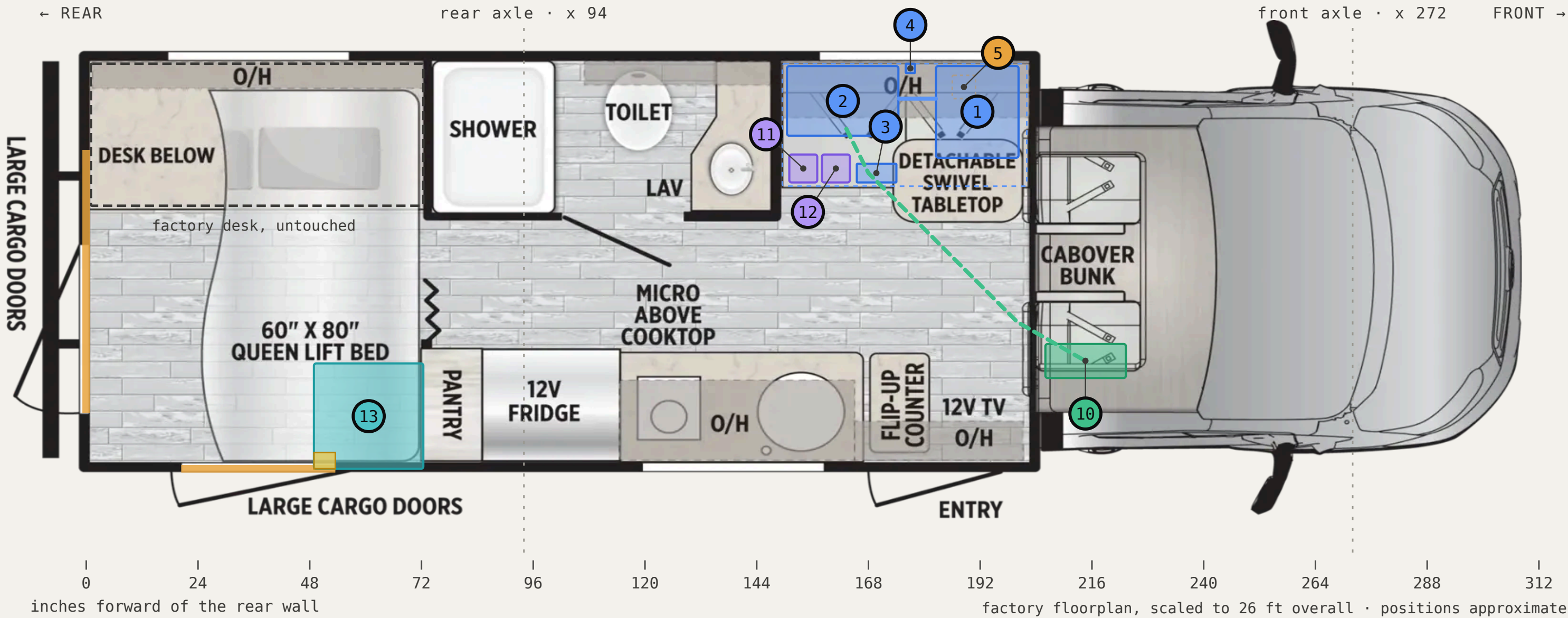
■ Washer-dryer

□ Stock coach

■ Fit check

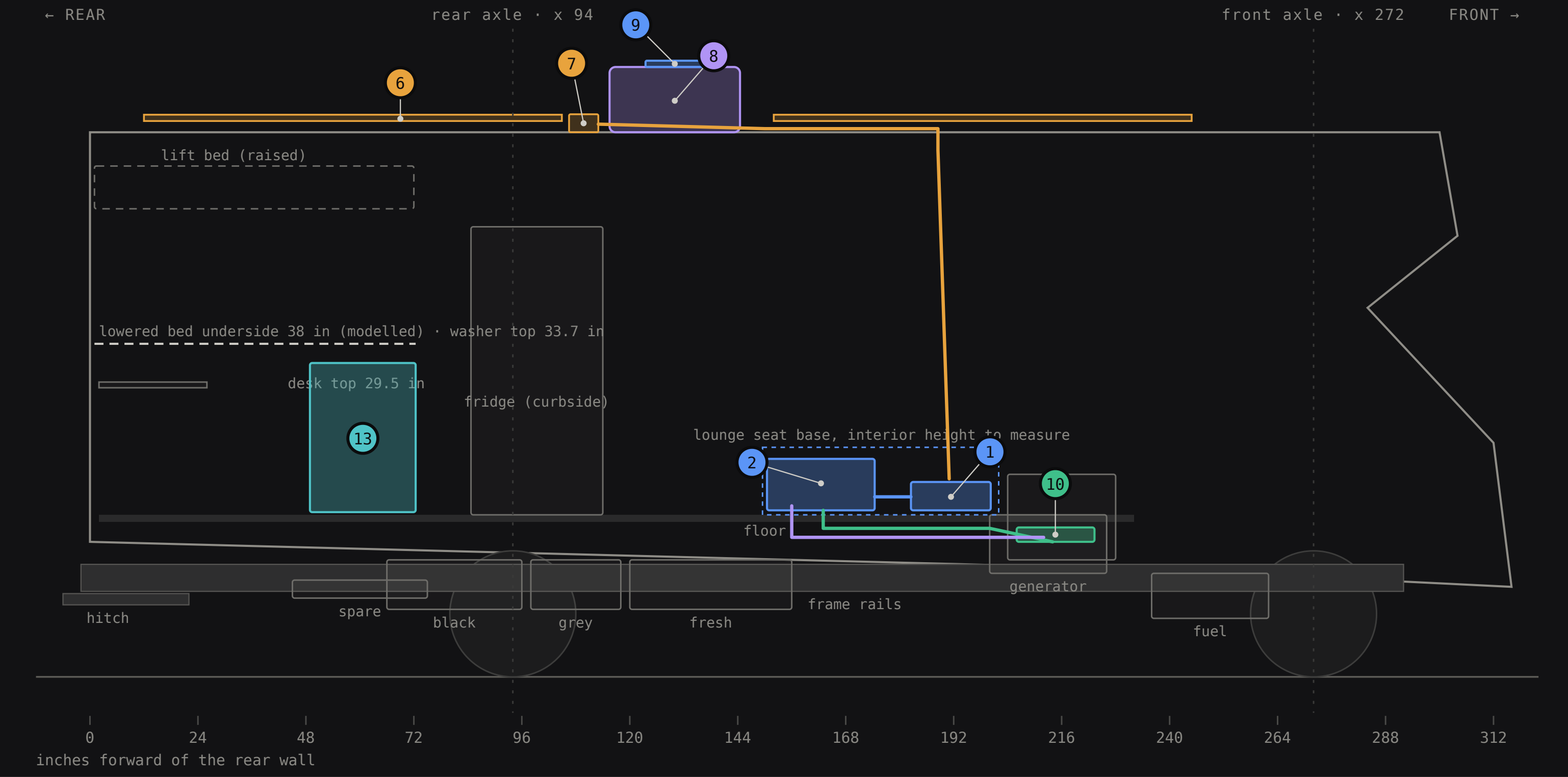
Floor plan

East to West's factory 22NF floorplan, scaled to 26 ft, with the new hardware placed on it.
Roadside (driver side) is at the top. Dashed shapes are above the ceiling or below the floor. Orange lines are the cargo door openings read off the floorplan.



Side section

A cut through the coach seen from the side, showing how high everything sits: roof, cabin and underbody.

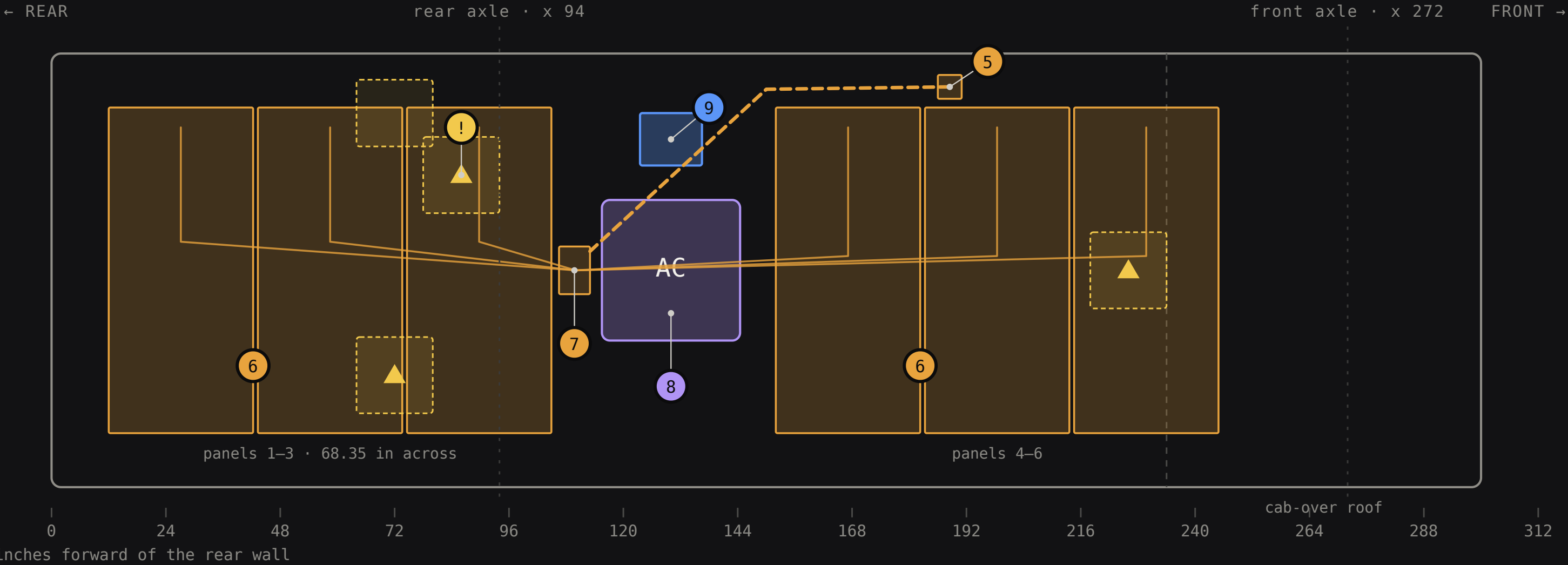


Underbody

VIEW 4

Roof

Six panels running across the roof, three behind the AC and three ahead, onto the cab-over. Yellow outlines mark the existing fans and skylight, which end up under the panels.



PLACEMENT SCHEDULE

Every new part, where it goes and what it connects to

#	PART · WHERE IT GOES	POSITION	TAKES THE PLACE OF	MOUNT AND ACCESS	CONNECTS TO
1	Bluetti RV5 hub Inside the driver-side lounge seat base, front end (your right when facing the seat from the aisle), lying flat against the outside wall.	x 182.5–200 in · 23.3–43 in roadside of centre · footprint 17.7 × 19.7 in, 6.3 in tall · centre about 92 in ahead of the rear axle	Storage in the front half of the lounge seat base.	Bolted to the floor through its mounting holes. 8 in gap to the battery stack. A louvred front panel on the seat base for airflow; the other faces sit closer than the 7.87 in the manual asks for (fit check).	48 V to the packs beside it (under 1 ft) · PV trunks from the roof gland directly above (about 12 ft each) · 120 V in from the transfer switch in the same base (about 2 ft) · 120 V out to the factory breaker panel (location to find) · 12 V out to the coach 12 V panel.

#	PART · WHERE IT GOES	POSITION	TAKES THE PLACE OF	MOUNT AND ACCESS	CONNECTS TO
2	2 × B4810, stacked Inside the same lounge seat base, rear end (your left when facing the seat), stacked two high against the outside wall.	x 150.5–174.4 in · 28–43 in roadside of centre · stack 23.94 × 14.96 × 11.42 in · centre about 68 in ahead of the rear axle	Storage in the rear half of the lounge seat base.	Strapped to a floor tray, each pack fused at its terminals. Needs at least about 12.5 in of clear height inside the seat base (11.42 in stack plus tray and strap).	48 V to the RV5 (under 1 ft) · alternator 48 V from the Orions under the passenger seat (about 6 ft, under the floor).
3	Main DC disconnect and PV breakers On the inside of the lounge base front panel, between the packs and the aisle, with the disconnect handle through the panel.	x 165.5–174 in · front panel	Front panel of the seat base.	Reachable from the aisle without lifting the seat or going under the coach.	Between the packs and the RV5, and between each PV trunk and its RV5 input.
4	Bluetti Epad (control screen) On the wall above the lounge, by the entry.	x 176–178 in · roadside wall · eye level	Wall space only.	Surface mount.	Data cable to #1 directly below (about 3 ft).
5	Roof cable gland Roof, driver-side edge, directly above the lounge, outboard of panels 4–6.	x 186–191 in · 36–41 in roadside of centre	Nothing. New sealed penetration.	Sealed gland with written roof-warranty coverage. The two PV trunks drop inside the driver-side wall into the lounge base.	Both PV trunks from #7 to #1.
6	6 × Callsun 275 W panels Roof, one crosswise row of six: panels 1–3 behind the AC, panels 4–6 ahead of it, running onto the cab-over roof.	x 12–105 and 152–245 in · each 68.35 in across the roof	Roof space only. The existing fans and skylight sit under the panels (see fit check).	Z-brackets or a low rail, about 2.5 in off the roof. Walk margin left on both sides.	Panel leads to #7 (4–12 ft each, about 48 ft total).
7	Combiner box (3 + 3, 25 A per panel) Roof centreline, between panels 1–3 and the AC.	x 106–113 in · centreline	Nothing.	Bolted to the roof, sealed. Two groups of three, each with a 25 A fuse per panel.	Panels in; two PV trunks out to #5 then #1 (about 12 ft each).
8	Furrion Chill Cube 18K (ducted) The existing 14 × 14 in AC opening. Swaps one-for-one with the factory unit.	Opening modelled at x 116–145 in (centre x 130). Real position unverified.	The factory AC (removed). Its ceiling assembly is replaced by the Chill Cube air distribution box, which feeds the existing ducts.	Bolts through the existing opening. Confirm the air distribution box mates to the factory ducts before buying.	120 V from the main panel (RV5 output).
9	Starlink Mini Roof, roadside of the AC on a flat mount.	x 124–137 in · 22–33 in roadside of centre	Roof space only.	Flat mount, adhesive or bolted. Cable rides the PV gland or the AC chase.	12 V from the RV5 12 V output.
10	2 × Victron Orion-Tr 12/48 Under the passenger (curbside) cab seat, close to the starter battery.	x 206–223 in · curbside · under the seat base	Space under the passenger seat. The cab seats swivel to face the table, so the swivel base must leave room (fit check).	Screwed to the seat base or the floor beside it. Ignition-switched. Fuse at the starter battery.	12 V from the starter battery (about 3 ft) · 48 V to the packs in the lounge base, under the floor (about 6 ft, about 16 A).
11	Transfer switch Inside the lounge base, on the inside of the front panel in front of the battery stack.	x 151–157 in · 18–24 in roadside of centre	Front-panel space in the seat base.	Screwed to the front panel. Service through the seat lid.	Generator in, under the floor (about 6 ft; generator bay unverified) · shore in through #12 · out to the RV5 AC input (about 2 ft).
12	Surge protector / EMS (hardwired) Beside the transfer switch on the lounge base front panel.	x 158–164 in · 18–24 in roadside of centre	Front-panel space in the seat base.	Screwed to the front panel next to #11.	Shore inlet in (inlet spot unverified; length depends on it) · out to #11.
13	Splendide WDV2200XCD (vented) Curbside corner at the forward end of the lift bed, backed against the curbside wall with its side against the pantry wall.	x 48.9–72.4 in · y 20.9–43.5 in curbside of centre · footprint 23½ W × 22⅝ D × 33⅞ H in (plus about 0.6 in plinth) · centre about 33 in behind the rear axle	Garage floor in that corner, under the forward end of the bed.	Bracket to the floor. Fits under the lowered bed only if the bed’s underside sits at least 33.7 in above the floor (modelled 38 in, unpublished). Vent out the curbside wall. Its rear 4½ in overlaps the forward end of the curbside cargo door opening (x 20–54), leaving about 29 in of the opening clear.	Hot and cold cross under the aisle from the bath (shower directly opposite, about 4 ft) or come back from the galley sink. Drain to the grey tank, which the model puts under the floor just ahead of this corner. Own 15 A circuit from the main panel.

Cable lengths are measured along the routes drawn in the 3D model, rounded to the nearest foot.

What the six-panel row needs, and what to do if it doesn't fit

LENGTH THE ROW NEEDS 233 in Six panels at 30.3 in, 1 in gaps, 10.5 in for the combiner before the AC, 29 in of AC, and 7.5 in after it. In the model it runs from x 12 to x 245.	ROOF THE MODEL USED ~300 in, estimate The model treats the roof as flat from the rear cap over the cab-over. The row starts 12 in from the rear cap and ends 11 in past the house-to-cab-over seam (x 234). The 1 in margin quoted earlier compared the row with the house section alone. Measure both sections.	MAXXFAN UNDER A PANEL No A MaxxFan Deluxe with rain dome stands 5.0 in tall closed and 9.1 in open. Panels on 2.5 in brackets can't pass over it.
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FALLBACKS, IN ORDER

- Tall rails over the fans.** Raise the panels so their underside sits about 10 in off the roof, clearing an open MaxxFan. Panel tops end up about 11.4 in high, still under the Chill Cube's 14.5 in, so the coach's overall height doesn't change. The cost is a heavier rack and more wind load.
- Leave a fan open.** Shift the row so a fan falls in a gap. Each uncovered fan costs one panel slot, leaving five panels: 1,375 W wired 3 + 2. Two panels in parallel need no per-panel fuses.
- Short roof.** Drop the front panel: five panels need about 202 in. Four panels (2 + 2, 1,100 W) need about 171 in.
- Tighten the gaps.** Move the combiner off the row to the roof edge to save about 6 in.

REAR AXLE CHECK

Axle loads added by the build, against the Transit 350 ratings

The coach's empty axle weights aren't published, so the table works backwards. It totals everything the build and a loaded trip add to each axle, then shows the heaviest the *empty* axle can weigh and still stay under its rating (front 4,630 lb, rear 7,275 lb). Included: the build at its modelled positions, a full fresh tank (290.5 lb at the tank), a 40 lb tow bar on the receiver, a 200 lb driver, and 350 lb of gear spread midship.

SCENARIO	ADDED TO FRONT AXLE	ADDED TO REAR AXLE	EMPTY REAR AXLE CAN BE AT MOST	EMPTY FRONT AXLE CAN BE AT MOST
Panels 29.8 lb each · gear midship	+434 lb	+1131 lb	6,144 lb	4,196 lb
Panels 29.8 lb each · gear in the rear garage	+257 lb	+1308 lb	5,967 lb	4,373 lb
Panels 65.7 lb each · gear midship	+476 lb	+1305 lb	5,970 lb	4,154 lb
Panels 65.7 lb each · gear in the rear garage	+299 lb	+1482 lb	5,793 lb	4,331 lb

- What it means.** With an empty coach around 9,000 lb, the rear axle stays legal if the empty rear axle weighs no more than about 5,790–6,140 lb, or roughly 64–68% of the empty weight on the rear. The front axle has room in every case: it can carry up to 4,150–4,370 lb empty. A CAT scale reading of the empty coach answers it.
- Gear placement matters as much as the box.** Putting the 350 lb of gear in the rear garage instead of midship adds another 177 lb to the rear axle. Every 100 lb moved from the garage to midship takes about

50 lb off the rear.

- **Why the lounge.** With the packs and RV5 in the lounge base, about 68–92 in ahead of the rear axle, the build puts about 213 lb less on the rear axle than with the packs in a tail box, and nothing hangs under the coach.

BATTERY CABLE

The 48 V run from the packs to the RV5 is under 1 ft

Both sit in the same lounge seat base, with an 8 in gap between the battery stack and the RV5. The pack fuse goes at the pack terminals. The alternator line from the Orions under the passenger seat is about 6 ft at about 16 A, and the PV trunks from the roof gland directly above are about 12 ft each.

FIT CHECKS

Nine things the tape measure has to confirm

1. **Lounge seat base.** Measure the inside of the driver-side lounge base: length (needs about 52 in for the stack, the 8 in gap and the RV5), depth from the wall (needs about 20 in for the RV5 plus the front-panel gear), and clear height (needs about 12.5 in for the stacked packs). Check what the factory put in there: furnace, water heater, tank or wiring.
2. **RV5 airflow.** The manual asks for 7.87 in of open air at the RV5. Inside the seat base it gets 8 in on the side facing the packs and a louvred front panel; the end wall and front gap are closer. Confirm which faces vent, and add a small fan if needed.
3. **Roof fans and skylight under the panels.** The panels sit 2.5 in off the roof, and the model puts two roof fans and the bath skylight under them. A MaxxFan Deluxe with rain dome is 23.2 × 16.6 in and stands 5.0 in tall closed and 9.1 in with the lid open (retailer spec, overall unit height). It cannot sit under a panel on 2.5 in brackets. See the roof fallback section.
4. **Roof length and AC position.** The row of six spans 233 in from the first panel edge to the last (x 12 to 245). The model treats the roof as flat from the rear cap over the cab-over to about x 300. Measure the flat roof length, whether the cab-over roof is flat and walkable, and the AC opening’s distance from the rear cap.
5. **Passenger seat swivel.** The cab seats swivel to face the table. Check what’s under the passenger swivel base and whether two Orions (each about 7.3 × 5.1 × 3.1 in) fit without blocking the swivel. Fallback: the floor beside the seat base.
6. **Generator bay, shore inlet and breaker panel.** Find the generator bay, the shore inlet and the factory breaker panel. The 120 V feeds come in under the floor to the lounge base; their lengths depend on where those are.
7. **Washer corner.** Measure the lift bed’s underside fully lowered at the forward curbside corner (needs at least 33.7 in), the floor space from the pantry wall back (needs 23½ in) and out from the curbside wall (needs 22¾ in), where the bath and galley supply lines run, and where the grey tank inlet is.
8. **Cargo doors.** The factory floorplan shows large cargo doors in the rear wall and on the curbside under the bed. The curbside door swings outward, so it still opens; the washer takes about 4½ in off the forward end of its roughly 33 in opening. Measure the opening.

3D REFERENCE

The same layout in the rendered model

